Enzo Porto Brasil

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EDUCATION

Feb 2024 – Jul 2026 **MSc in Statistics**
Graduate Program in Statistics, University of Brasília (UnB) — Brasília, Brazil

THESIS *"Statistical Modelling of Spatial Extremes in Complex Systems"*
Developed a Bayesian hierarchical geostatistical model for spatial extremes with BGEV marginals, allowing all distributional parameters to vary spatially through latent Gaussian structures. The model is designed to capture bimodality, tail behaviour, and spatial heterogeneity within a unified probabilistic framework.

METHODS Bayesian hierarchical modelling; Extreme Value Theory; latent Gaussian processes with spatial covariance; support-preserving parameter transformations; prior specification and regularisation; non-centred parameterisation; HMC/NUTS inference in Stan; spatial confounding control via QR projection; MCMC simulation and diagnostics; parameter recovery; uncertainty quantification; and estimation of extreme quantiles.

RESEARCH Alongside the dissertation, developed independent research in statistical methodology, extreme value theory, climatology, and astrostatistics, resulting in 4 submitted manuscripts. 2 further manuscripts are being developed from the thesis itself, one methodological and one applied. Additional research conducted during the MSc led to 3 oral/invited presentations and 6 scientific posters at national and international events.

COURSES Statistical Inference, Bayesian Inference, Mathematical Statistics, Computational Techniques in Statistics, Machine Learning, Geostatistics, Extreme Value Theory and Copulas, Generalized Linear Models.

STATUS Thesis defended. All academic requirements completed. Formal diploma issuance pending.

Aug 2018 – Dec 2023 **BSc in Statistics**
University of Brasília (UnB) — Brasília, Brazil

THESIS *"Imputation Methods for Missing Data Using Predictive Modelling"*
Monte Carlo simulation study comparing single and multiple imputation strategies, analysing regression and machine learning methods across different missing data scenarios.

TRAINING Probability and statistical inference; regression; Bayesian methods; computational statistics; stochastic processes; time series; multivariate analysis; computing; experimental design; calculus; numerical methods; linear algebra; fundamentals of physics; fundamentals of astronomy and astrophysics.

RESEARCH CNPq-funded undergraduate research project on *"Imputation Methods for Dichotomous Variables"*. Research on simple and multiple imputation methods through simulation and real data analysis in R.

Sep 2022 – Mar 2023 **Academic Exchange on Modelling and Data Science**
Lodz University of Technology — Łódź, Poland

On campus coursework in statistical physics, experimental physics, stochastic processes, computational modelling, computer science, and logic, completed in partnership with the University of Brasília.

Jun 2016 – Jun 2017 **Building Electrician: Control and Industrial Processes**
Professional Education Center – Brasília Technical School (CEP–ETB) — Brasília, Brazil

Completed intermediate professional qualification within the Technical Programme in Electrotechnics.

RESEARCH OUTPUTS

SUBMITTED MANUSCRIPTS AND PREPRINTS

[1] **Extremes of Solar Spectral Irradiance in the SORCE/XPS Record**
Brasil, E.; Otiniano, C.; Brito, C.; Albernaz, B.; Morales, F.
Accepted for publication in *Frontiers in Astronomy and Space Sciences*, Astrostatistics topic.
Article (Frontiers): doi:10.3389/fspas.2026.1970861 Preprint [astro-ph.SR]: arXiv:2608.30878 Code (Zenodo): doi:10.5281/zenodo.23269103

[2] **Multiple Imputation Methods under Extreme Values**
Brasil, E.
Submitted to the *Austrian Journal of Statistics*.
Preprint [stat.CO]: arXiv:2602.04751

[3] **Joint return levels of maximum temperature and minimum relative humidity by combining copulas with an extreme value framework for bimodal data**
Albernaz, B.; Otiniano, C.; Brito, C.; Morales, F.; Brasil, E.
Submitted to *Advances in Statistical Climatology, Meteorology and Oceanography*.
Preprint [stat.ME]: arXiv:2608.10252

[4] **Conditional copula representations and extremal bounds for multivariate statistical functionals**
Vila, R.; Otiniano, C.; Brito, C.; Brasil, E.
Submitted to *Statistics and Probability Letters*.
Preprint [math.ST]: arXiv:2607.26256

CORE METHODS, SCIENTIFIC COMPUTING & DATA

Statistical inference, uncertainty & hierarchical modelling
Bayesian and frequentist inference; hierarchical and latent-variable models; Gaussian processes; parameter inference; uncertainty quantification; model criticism and diagnostics.

Simulation, calibration & model assessment
Monte Carlo simulation; simulation studies; synthetic-data experiments; parameter recovery; calibration and sensitivity analysis; posterior and MCMC diagnostics; robustness assessment.

Astrostatistical & physical data analysis
astrostatistical applications; NASA SORCE solar spectral irradiance; space science and time indexed physical measurements; heterogeneous observational datasets; data validation; uncertainty assessment.

Dependence, extremes & complex stochastic structure
Extreme Value Theory; GEV/BGEV models; copulas; multivariate and extremal dependence; spatial and spatiotemporal modelling; covariance structures.

Scientific & Bayesian computing
R/RStudio/R Markdown for statistical modelling and simulation; Python for scientific computing, data analysis, and automation; Stan for hierarchical Bayesian inference with HMC/NUTS.

Climate & atmospheric systems
Solar irradiance and air density; temperature and humidity extremes; climate and financial risk.

Quantitative social & public-sector research
Education indicators, national microdata and longitudinal records; public policy; diversity, equity and inclusion; administrative data validation.

Reproducible data workflows
SQL, DBeaver, ODBC; LaTeX; Git/GitHub; reproducible reporting; Shiny and Power BI; Stata and SAS.

RESEARCH & PROFESSIONAL EXPERIENCE

Sep 2025 – Present
Statistician
HU Brasil, Federal University Hospitals — Brazil
- Analysis and creation of large observational and administrative datasets within the Brazilian public sector, with emphasis on data quality, validation, robustness, and reproducible research.
- Develop automated analytical pipelines, dashboards, and research outputs; contribute to published and ongoing studies on diversity, equity and inclusion, social impacts, and psychological research.

May 2025 – May 2026
Senior Research Assistant and Data Science Fellow
Ipea, Institute for Applied Economic Research — Brazil
- Conducted qualitative and quantitative research with large national microdata, longitudinal datasets, and territorial information, integrating and validating heterogeneous sources across spatial and temporal scales.
- Developed reproducible statistical workflows, automated analyses, and dashboards; coauthored three technical research notes and a forthcoming book chapter on Brazilian education, with findings communicated through scientific events.

Aug 2024 – Dec 2025
Technical Officer, International Cooperation Mercosul
Inep, National Institute for Educational Studies and Research — Brazil
- Supported South American academic cooperation and undergraduate programme evaluation within the Mercosul accreditation system, including liaison with Latin American partners in meetings conducted in Portuguese and Spanish.

Sep 2023 – May 2025
Data Analyst and Consultant
Alcance Profissional — Brazil
- Built automated data workflows, reproducible analytical pipelines, and reporting processes, with focus on data organization, efficiency, and repeatability.

Oct 2021 – Sep 2022
Statistics Intern
FGCoop, Brazilian Credit Cooperative Guarantee Fund — Brazil
- Supported statistical monitoring, indicator analysis, risk modelling activities, and quantitative reporting.

FELLOWSHIPS, SCHOLARSHIPS & DISTINCTIONS

NSF EPSCoR International Research Travel Support
International research support. Supported by NSF EPSCoR and the Center for Space Plasma and Aeronomic Research (CSPAR), The University of Alabama in Huntsville, for an oral astrostatistics presentation at UAH and subsequent publication support for the associated *Frontiers in Astronomy and Space Sciences* manuscript.

CNPq MSc fellowship and Admission Distinction
Ranked 1st in MSc admission. First among 20 admitted candidates to the MSc in Statistics at UnB; awarded a two-year CNPq graduate scholarship.

Ipea National Research Fellowship
Ranked 1st for the project. Selected for Ipea's data science research team through a nationwide competitive selection (12 months).

CNPq and UnB Undergraduate Research Support
Undergraduate research funding. Institutional Scientific Initiation Scholarship Programme and academic teaching assistantship support (18 months).

PPGEST/UnB Scientific Travel Funding
Competitive scientific travel support. Awarded for research presentations at national and international scientific events from 2024 to 2026.

SCIENTIFIC PRESENTATIONS

Jun 2026 — Invited speaker
Extremes in Nature: statistics, uncertainty, and forecasting in the face of extreme phenomena
1st Workshop of the Graduate Program in Statistics, University of Brasília, Brazil

Nov 2025 — Oral
Joint extremes of solar irradiance and air density: an EVT and copula approach to tail uncertainty
Systematic and Measurement Errors across the Sciences: Astrostatistics and Data Science, The University of Alabama in Huntsville, USA; astrophysics applications track

Oct 2025 — Poster
Analysis of extreme solar irradiance in NASA's Solar Radiation and Climate Experiment (SORCE) mission: an Extreme Value Theory approach
5th Paraíba Meeting of Statistics, João Pessoa, Brazil

Oct 2025 — Poster
Impact of extreme values on multiple imputation of missing data using MICE in R
19th School of Regression Models, João Pessoa, Brazil

Aug 2025 — Poster
Climate-financial risk modelling for the agricultural sector using bivariate copulas
69th Brazilian Statistics Meeting (RBras) and 21st SEAGRO, Vitória, Brazil

Aug 2025 — Poster
Bivariate analysis of maximum temperature and minimum relative humidity in Brasília (DF)
69th Brazilian Statistics Meeting (RBras) and 21st SEAGRO, Vitória, Brazil

Aug 2025 — Poster
Extreme solar irradiance and its relationship with atmospheric variables: an approach using Extreme Value Theory and copulas
69th Brazilian Statistics Meeting (RBras) and 21st SEAGRO, Vitória, Brazil

Sep 2024 — Oral
Assessing imputation methods for missing data using statistical models: a simulation approach
1st Brazil and Colombia Binational Colloquium on Probability and Statistics, Natal, Brazil

Sep 2022 — Oral
Imputation techniques for dichotomous variables
University of Brasília Undergraduate Research Congress, Brasília, Brazil

Nov 2026 — Scheduled Poster
Astrostatistical inference for extremes of solar spectral irradiance in the SORCE/XPS record
XXVIII Ciclo de Cursos Especiais in Astronomy and Astrophysics, Observatório Nacional, Rio de Janeiro, Brazil

2026 — Accepted Poster
A Bayesian hierarchical model for spatial extremes with bimodality
Brazilian Symposium on Probability and Statistics (SINAPE)

2026 Accepted Poster **Listening to the workforce: statistical planning of the Diversity, Equity, and Inclusion Census in Brazil's largest public network of university hospitals**
Brazilian Symposium on Probability and Statistics (SINAPE)

2026 Accepted Oral **The expansion of full-time education in Brazil (2007–2024): federal programmes, funding, and patterns of enrolment growth**
2nd Pan-American Meeting on the Economics of Education

2026 Accepted Oral **Full-time education in Brazil: funding, expansion, and challenges under the Fundeb framework**
50th Annual Meeting of ANPOCS

ADDITIONAL RESEARCH OUTPUTS

MANUSCRIPTS IN PREPARATION

[5] **Characterising complex dependence between extreme solar irradiance and air density**
First-author manuscript in preparation; submission planned for Nov. 2026.

[6] **Bayesian hierarchical modelling of spatial extremes with BGEV marginals**
First-author manuscript in preparation; submission planned for Jan. 2027.

INTERDISCIPLINARY CONTRIBUTIONS

[7] **Full-Time Education in Brazil: Financing, Expansion, and Challenges in Light of Fundeb**
Xavier, A.; Senkevics, A.; Brasil, E.; Rocha, V.
Book chapter in *Novo Fundeb: balanço dos primeiros anos de implementação (2021–2026)*, 2026; in Portuguese.
Forthcoming; book launch in December 2026.

[8] **Labor Relations at the Brazilian Hospital Services Company: Democratization and the Promotion of Diversity, Equity, and Inclusion**
Dau, D.; Junqueira, E.; Brasil, E.; Lobo, L.; Viana, L.; Maia, M.; Sousa, T.
Revista Jurídica da Empresa Brasileira de Serviços Hospitalares, 2(2), 2025; published in Portuguese.

[9] **Full-Time Education Dashboard: Data Sources, Implementation, and Analytical Use**
Brasil, E.; Resende, M.; Nascimento, P. M.
Technical research note, Ipea, 2026; in Portuguese.
Editorial review.

[10] **Calculation and Presentation of Educational Indicators in the 2024–2027 Multi-Year Plan**
Brasil, E.; Candeia, L.; Rocha, V.; Senkevics, A.
Technical research note, Ipea, 2026; in Portuguese.
Editorial review.

[11] **The Evolution of Full-Time Education in Brazil**
Resende, M.; Santos, F.; Brasil, E.; Brito, A.
Technical research note, Ipea, 2026; in Portuguese.
Editorial review.

COURSES & ADDITIONAL TRAINING

Statistics, Probability & Mathematical Methods

- Statistical Modelling of Digital Images *19th School of Regression Models (EMR) · João Pessoa* (Oct 2025)
- 1st Workshop on Neural Simulation-Based Inference *The Statistical Methods for the Physical Sciences · Carnegie Mellon University* (Oct 2025)
- Missing Data in Longitudinal Analysis *25th SINAPE · Fortaleza* (Aug 2024)
- Univariate and Multivariate Linear Regression Models for Heavy-Tailed, Symmetric and Asymmetric Data *17th School of Regression Models (EMR) · Brazilian Institute of Geography and Statistics (IBGE)* (Dec 2021)
- Basic Algebra *National Autonomous University of Mexico (UNAM) · Coursera* (Jul 2020)
- Calculus: Quadratic Model *Monterrey Institute of Technology and Higher Education (ITESM) · Coursera* (Jul 2020)
- Calculus: Other Models *Monterrey Institute of Technology and Higher Education (ITESM) · Coursera* (Jul 2020)
- Calculus: Linear Model *Monterrey Institute of Technology and Higher Education (ITESM) · Coursera* (Jul 2020)
- Calculus: Cubic Model *Monterrey Institute of Technology and Higher Education (ITESM) · Coursera* (Jul 2020)
- Probability and Statistics: To p or not to p? *University of London · Coursera* (Jul 2020)
- Mathematical Foundations for Cryptography *University of Colorado System · Coursera* (Jul 2020)
- Fibonacci Numbers and the Golden Ratio *The Hong Kong University of Science and Technology · Coursera* (Jul 2020)

Data Science & Scientific Computing

- Algorithm Programming in Python *Brazilian Space Agency (AEB)* (Oct 2026)
- R and Python with Spark for big-data *5th Paraíba Meeting of Statistics (EPBEST)* (Oct 2025)
- Machine Learning with `tidymodels` in R *17th School of Regression Models (EMR) · Brazilian Institute of Geography and Statistics (IBGE)* (Dec 2021)
- Introduction to the R Language *University of Brasília (UnB)* (Oct 2021)
- Python for Beginners *University of Brasília (UnB)* (Oct 2021)
- Intermediate Python *DataCamp* (Sep 2021)
- Introduction to Python *DataCamp* (Sep 2021)
- Reporting with R Markdown *DataCamp* (Sep 2021)
- Cleaning Data in R *DataCamp* (Aug 2021)
- Introduction to the Tidyverse *DataCamp* (Aug 2021)
- Intermediate R *DataCamp* (Aug 2021)
- Introduction to R *DataCamp* (Aug 2021)

Astronomy, Space, Physical & Earth Sciences

- The Nature of Dark Energy · Pedro Ferreira (University of Oxford) *XXVIII Ciclo de Cursos Especiais (CCE) · Observatório Nacional* (Nov 2026 · scheduled)
- Galaxy Evolution in Clusters · Bianca Poggianti (INAF, Padova) *XXVIII Ciclo de Cursos Especiais (CCE) · Observatório Nacional* (Nov 2026 · scheduled)
- Extended Bodies in the Solar System: Rotation, Tides, and Figures · Nicolas Rambaux (Paris Observatory) *XXVIII Ciclo de Cursos Especiais (CCE) · Observatório Nacional* (Nov 2026 · scheduled)
- The Milky Way after Gaia · Victor Debattista (Lancashire University) *XXVIII Ciclo de Cursos Especiais (CCE) · Observatório Nacional* (Nov 2026 · scheduled)
- Remote Sensing *Brazilian Space Agency (AEB) · National Institute for Space Research (INPE) · Federal University of São Carlos (UFSCar)* (Oct 2026)
- Astronomy *Brazilian Space Agency (AEB) · Federal University of Rio Grande do Norte (UFRN)* (Sep 2026)
- Satellites *Brazilian Space Agency (AEB) · Federal University of Rio Grande do Norte (UFRN)* (Sep 2026)
- Introduction to Physical Oceanography *Brazilian Space Agency (AEB)* (Mar 2026 · in progress)
- Climate Change and Sovereign Risk *Asian Development Bank Institute* (Feb 2025)
- Quantum and Statistical Physics Workshop *International Center of Physics · University of Brasília (UnB)* (Apr 2024)
- AstroTech: The Science and Technology behind Astronomical Discovery *University of Edinburgh · Coursera* (Jul 2020)

- Quantum Mechanics *École Polytechnique · Coursera* (Jul 2020)
- Philosophy and the Sciences: Introduction to the Philosophy of Physical Sciences *University of Edinburgh · Coursera* (Jul 2020)
- Origins of Life in the Cosmic Context *University of São Paulo (USP) · Coursera* (Jul 2020)
- Fundamentals of Astronomy and Astrophysics *University of Brasília (UnB)* (2020)

Research Methods & Academic Practice
- Research Data Management *Elsevier Researcher Academy* (Aug 2025)
- Research Collaborations *Elsevier Researcher Academy* (Aug 2025)
- Research Integrity: Publication Ethics *Nature Masterclasses · Springer Nature* (Aug 2025)
- Creative Writing for Scientific Articles *University of Brasília (UnB)* (Sep 2020)

Technical & Professional Training
- Lean Six Sigma White Belt *MF Training · Marcelo Machado Fernandes* (Apr 2021)

Social Sciences, Education & Humanities
- (Re)learning to Study: Meaning and Autonomy *University of Brasília (UnB)* (Oct 2021)
- Eureka! Creativity in Theory and Practice *University of Brasília (UnB)* (Sep 2020)
- Learning Primary Mathematics *University of the Andes · Coursera* (Jul 2020)
- Primary Mathematics Content *University of the Andes · Coursera* (Jul 2020)
- Skepticism *University of California, Irvine · Coursera* (Jul 2020)
- The Science of Well-Being *Yale University · Coursera* (Jul 2020)
- Arts and Heritage Management *Bocconi University · Coursera* (Jul 2020)
- Big History: From the Big Bang until Today *University of Amsterdam · Coursera* (Jul 2020)
- Introduction to Philosophy *University of Edinburgh · Coursera* (Jul 2020)

ACADEMIC SERVICE, TEACHING & VOLUNTEERING

Academic leadership & service

Jun 2026 **Organizing committee member** 1st Extension Forum, Institute of Exact Sciences, University of Brasília
2025 **Peer reviewer (6 manuscripts)** *Computational Statistics*, Springer Nature
2024 **Academic event monitor** 7th Workshop on Survival Analysis and Applications, University of Brasília
Jan 2022 **General coordinator** 50th Summer School and 14th Summer Workshop in Mathematics, University of Brasília
May 2021 **Academic event monitor** 6th Colloquium on Mathematics of the Central-West Region, University of Brasília

Teaching & academic support

Mar–Jul 2025 **Graduate Teaching Internship in Statistics** University of Brasília
2019 **Teaching Assistantship in Probability Theory** University of Brasília
Long-term **Academic Tutoring and Mentoring** Secondary school and undergraduate students on statistics, mathematics, and physics

LANGUAGES

Portuguese: native. **English**: fluent (TOEFL iBT: 5.0, C1). **French**: intermediate. **Spanish**: reading and listening comprehension. **Polish**: introductory exposure.